

# **Predicting Formal Protest Petitions Against Housing Development**

Evidence from Austin, Texas

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April 16, 2026

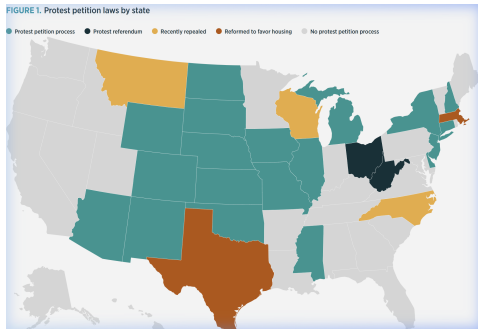
# Austin's Housing Crisis and Localized Opposition

## The Housing Affordability Problem

- Austin population grew  $\approx 40\%$  between 2010–2024, intensifying supply constraints.
- New housing requires *discretionary* rezoning approval — not a by-right process.
- Individual projects face public hearings, council votes, and organized opposition.

## The Political Mechanism

Texas law gave adjacent property owners a formal institutional veto tool: the **protest petition**.



# The Protest Petition: An Institutional Veto

## Texas LGC Chapter 211 — The 20% Rule

- Owners of **20% of land area** within 200ft of a proposed rezoning may file written protest.
- A valid petition forces a **supermajority vote** ( $\frac{3}{4}$  of council) instead of a simple majority.
- On Austin's 7-member council: 4 votes → 6 votes needed for approval.
- 21 states have similar architectures (Furth & McKinley 2025).

*A localized minority of adjacent owners can effectively block citywide housing decisions.*

| PETITION                                |                                               |                    |                           |               |
|-----------------------------------------|-----------------------------------------------|--------------------|---------------------------|---------------|
| Case Number:                            | C14-2007-0131                                 |                    | Date:                     | July 29, 2008 |
|                                         | 1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST |                    |                           |               |
| Total Area Within 200' of Subject Tract |                                               |                    |                           | 293,002.55    |
| 1                                       | 02-0906-0502                                  | MONAHAN CASEY J    | 7423.26                   | 2.53%         |
| 2                                       | 02-0906-0503                                  | MONAHAN CASEY J    | 7390.73                   | 2.52%         |
| 3                                       | 02-0906-0504                                  | CLARK MICHAEL G &  | 7354.85                   | 2.51%         |
| 4                                       | 02-0906-0506                                  | RITA S DE BE       | 1645.43                   | 0.56%         |
|                                         |                                               | MARTINSON KATHY L  |                           |               |
|                                         |                                               | CHILES ROSALIE     |                           |               |
| 5                                       | 02-0906-0507                                  | BENSON             | 1618.17                   | 0.55%         |
| 6                                       | 02-0906-0508                                  | KOCHERT KELLEY     | 1612.79                   | 0.55%         |
| 7                                       | 02-0906-0509                                  | MONAHAN CASEY J    | 1419.72                   | 0.48%         |
| 8                                       | 02-0906-0606                                  | TIMMERMANN TERRELL | 3452.33                   | 1.18%         |
|                                         |                                               | MERCADO FREDDY G   |                           |               |
| 9                                       | 02-0906-0607                                  | & MARIA R          | 9482.82                   | 3.24%         |
| 10                                      | 02-0906-0901                                  | LEGETT GEORGIA F   | 4459.36                   | 1.52%         |
| 11                                      | 02-0906-0911                                  | LEGETT GEORGIA F   | 7856.05                   | 2.68%         |
|                                         |                                               | MEDINA JAMES &     |                           |               |
| 12                                      | 02-0906-1001                                  | KRISTINE M GARA    | 13204.88                  | 4.51%         |
| 13                                      | 02-0906-1011                                  | RECER DANALYNN     | 9454.11                   | 3.23%         |
| 14                                      | 02-0906-1013                                  | BRINSMADE LOUISA C | 7634.37                   | 2.61%         |
| 15                                      |                                               |                    |                           | 0.00%         |
| 16                                      |                                               |                    |                           | 0.00%         |
| 17                                      |                                               |                    |                           | 0.00%         |
| 18                                      |                                               |                    |                           | 0.00%         |
| 19                                      |                                               |                    |                           | 0.00%         |
| 20                                      |                                               |                    |                           | 0.00%         |
| 21                                      |                                               |                    |                           | 0.00%         |
| 22                                      |                                               |                    |                           | 0.00%         |
| 23                                      |                                               |                    |                           | 0.00%         |
| 24                                      |                                               |                    |                           | 0.00%         |
| 25                                      |                                               |                    |                           | 0.00%         |
| 26                                      |                                               |                    |                           | 0.00%         |
| 27                                      |                                               |                    |                           | 0.00%         |
| Validated By:                           | Stacy Meeks                                   |                    | Total Area of Petitioner: | 84,008.88     |
|                                         |                                               |                    | Total %                   | 28.67%        |

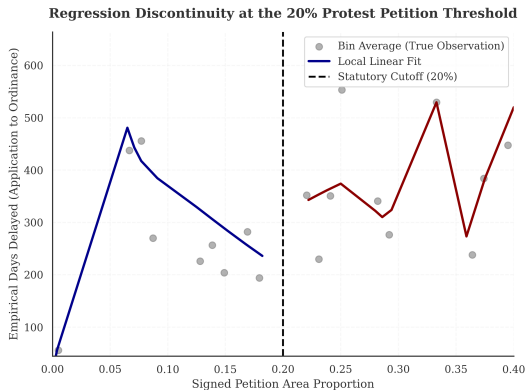
# Why This Outcome Matters: Petitions Impose Real Institutional Costs

## Reduced-Form Threshold Discontinuity at 20%

- **Running variable:** Reconstructed petition-share (% signed adjacent area within buffer).
- **Outcome:** Council processing time (filing to final vote).
- **Estimator:** Triangular-kernel WLS at the statutory 20% margin.

**Result:** Crossing the threshold is associated with a **42.5-day increase** (6.1 weeks) in processing time.  
95% CI: [26.4, 58.6] days.

*Robustness: McCrary density test  $p > 0.05$ ; symmetric*



# The Research Question

## The Prediction Problem

Can we anticipate formal protest petitions *before* public hearings, using only filing-date administrative records?

## Filing-Date Features (observable at application)

- Site geometry: lot area, height/FAR deltas.
- Current vs. requested zoning class.
- Buffer demographics and TCAD appraisal aggregates.
- Historical petition rates in the council district.

## A Sample Case

| Feature         | Case C14-2018-0156    |
|-----------------|-----------------------|
| Gross Site Area | 4.92 Acres            |
| Height Request  | +15.5 ft              |
| FAR Request     | +0.43                 |
| Neighborhood    | Riverside             |
| <b>Result</b>   | <b>Valid Petition</b> |

*Sensitive attributes (race, ethnicity) are **excluded** from predictive features and reserved for post-hoc equity audits only.*

# The Rarity Problem

## Sample Constraints

- **Universe:** 2007–2024 Austin discretionary rezonings.
- **Analytic sample:**  $N = 6783$  cases after exclusions.
- **Petition base rate:**  $\approx 2.6\%$  — fewer than 1 in 38 cases.
- **Test-set positives:** Only  $n_{\text{pos}} = 15$  petitioned cases in the temporal holdout.

Sparse positives widen confidence intervals and prevent reliable case-level probability use.

The appropriate output is a **relative risk ranking**, not a probability score for individual decisions.

## Evaluation: Strictly Temporal Holdouts

- No random train/test splits.
- Train on years  $\leq t$ ; evaluate on years  $> t$ .
- Mimics forward administrative use.

## Primary metric: Precision-Recall AUC

AUROC is omitted: with  $\approx 97.4\%$  true negatives, the large denominator artificially inflates the curve and obscures minority-class performance. PR-AUC forces the model to balance recall against the precision cost of false alarms.

# Timeline Integrity: As-Of Feature Construction

## The Engineering Challenge

- **Data Recovery:** 15+ years of zoning agendas scraped and parsed from PDF clerk records.
- **Milestone Reconstruction:** Municipal IT overwrites historical timestamps; filing dates back-calculated from statutory processing minimums.
- **No Future Leakage:** ACS releases, TCAD appraisal rolls, and petition records anchored to their publication date, not filing date.
- **Qualitative Grounding:**  $N = 8$  semi-structured interviews (IRB Protocol

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### Research Project Overview

*Predicting NIMBYism in Austin's Housing Reform Era*

#### Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

#### Study Objectives

**Identify Patterns:** Characterize the opposition rhetoric and arguments used in process petitions to oppose rezoning and identify demographic predictors of this opposition.

**Evaluate Predictive Tools:** Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

**Democratic Implications:** Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

#### Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

**Activity:** A 45-60 minute one-on-one interview conducted virtually or in person.

**Topics:** Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

#### Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol YU1800.

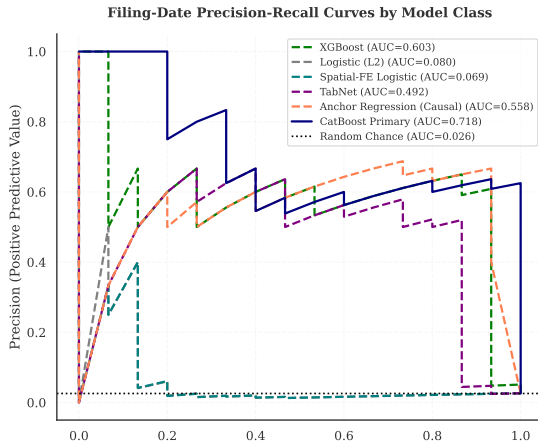
Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

# Predicting Resistance: Filing-Date Results

Tree Ensembles Dominate at the Top of the Risk Ranking





# Operational Utility: What the Triage Actually Looks Like

## Top-Decile Risk Ranking

- Flag the **highest-risk 10%** of incoming cases.
- Precision in that tier: **25.4%** (CI: 16.1%–37.8%).
- Lift over base rate: **9.9x**.
- True positives captured: 15 of 59 flagged cases.

**In plain terms:** A planner reviewing the top-decile flagged cases is nearly **10×** more likely to encounter a genuine petition case than by random case selection.

## The Appropriate Use Case

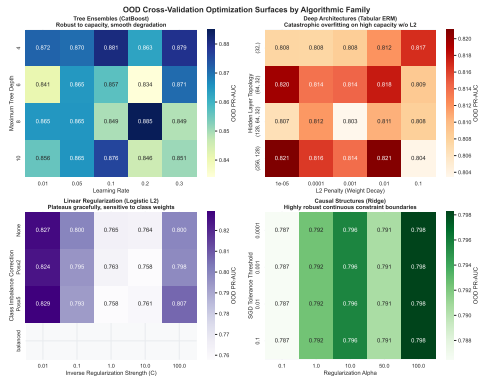
- ✓ **Triage:** Prioritize outreach and early negotiation for high-risk cases.
- ✓ **Monitoring:** Flag cases that historically co-occur with petition mobilization.
- × **Not:** Deterministic probability thresholds for case-level go/no-go decisions.
- × **Not:** Developer site-screening to avoid high-risk neighborhoods.

*Ethics Warning: This model ranks risk. It should not be used to pre-screen sites for developers seeking the path of least resistance.*

# Modeling Architectures: Four-Family Benchmark

## Pre-Registered Architecture Families

- **Tree Ensembles (CatBoost):** Robust to feature scale diversity; dominant in tabular planning data.
- **Deep Architectures (TabNet/MLP):** Sensitive to tuning; catastrophic overfitting without heavy L2 regularization.
- **Regularized Linear (Logistic L2):** Competitive at T+0 horizons; degrades sharply over time.
- **Distributionally Robust (V-REx):** Anchor regression; causal-stability baseline.



All tuned to maximize OOD PR-AUC under strict

# Dealing with Collinearity: Semantic Feature Clustering

## The Problem

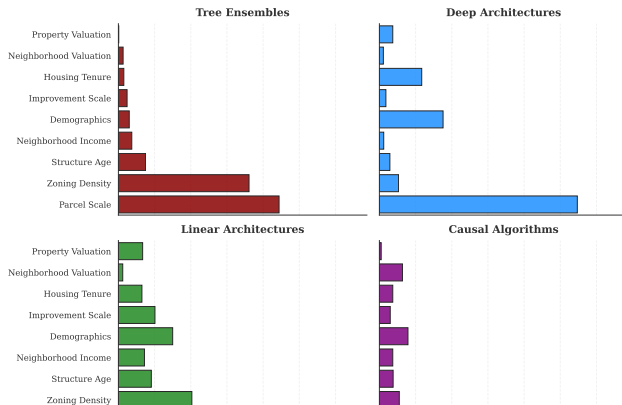
Administrative real estate records contain highly correlated proxies: site area, deed acreage, and structure square footage all measure the same physical concept.

Standard SHAP attribution dilutes importance arbitrarily across collinear variables.

## The Solution

Group variables into semantic clusters (Spearman  $|r| > 0.70$ ), then **sum absolute TreeSHAP values** within each cluster

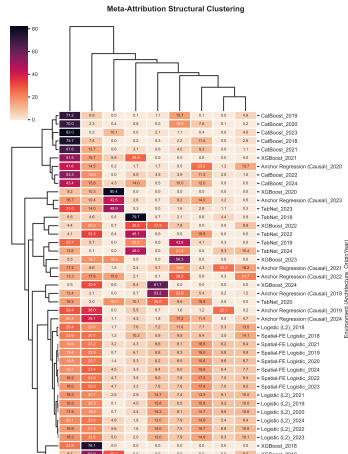
Comparative Primary Reliance: Top Feature Clusters Across Architectures



# Architecture Determines Which Features Drive Predictions

## Key Finding: Architecture-Dependent Feature Reliance

- **Tree models** allocate  $>80\%$  of predictive weight to *socio-demographic* clusters (Demographics, Income, Tenure).
- **Deep models** shift toward *physical/geographic* attributes (Parcel Scale, Valuation, Structure Age).
- Same data, same task, same outcome — different internal logic.



# Temporal Drift: Predictive Performance Over Time

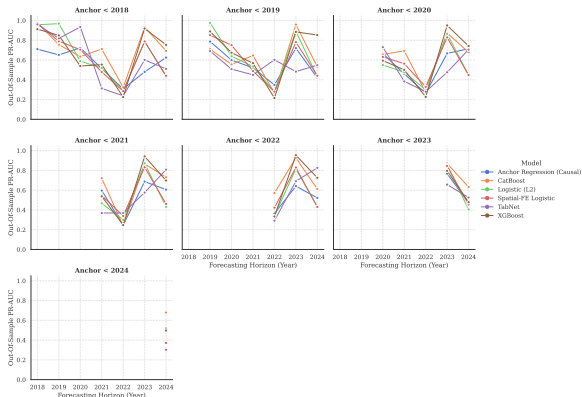
## When Does the Model Break?

- Models trained on historical data encounter *concept drift* when the institutional environment changes.
- **Linear models** collapse at long forecasting horizons (T+3 and beyond).
- **Tree ensembles** degrade more gracefully: dominant in 15 of 16 volatile-anchor evaluation cells.

CatBoost:  $T+0$  (2019) = 0.659  $\rightarrow$   $T+3$  (2022) = 0.329  $\rightarrow$   $T+5$  (2024) = 0.885.

This dip-and-recovery is treated as **descriptive out-of-sample variation** — no

Temporal Predictive Drift by Model Architecture (PR-AUC)

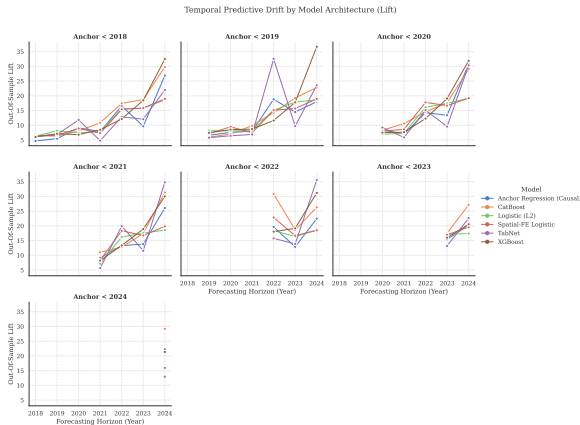


# Signal Dilution: When Retraining Hurts

## The Paradox

- When petition rates fall post-ordinance, absolute PR-AUC declines even if ranking is perfect. **Lift** ( $\text{PR-AUC} \div \text{base rate}$ ) corrects for this.
- Models trained on *pre-2018 data* achieve  $\sim 34\times$  lift in 2024.
- Retraining on modern data *degrades* lift to  $\sim 19\times$ .

**Why:** Post-ordinance labels starve the model of positive examples. The frozen legacy model preserves spatial sorting intelligence learned when the institutional environment was active.



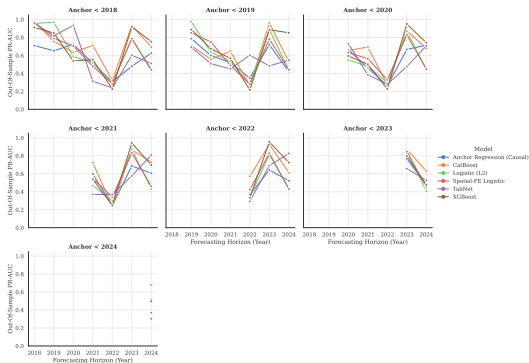
# Agency in Temporal Shifts: What the 2022 Dip Tells Us

## The 2022 Performance Pattern

- **Finding:** PR-AUC drops sharply across all architectures in the 2022 evaluation window.
- **Interpretation:** Consistent with pandemic-era filing disruptions (2020–2021) reaching the evaluation window, combined with administrative uncertainty during the Land Development Code revision period.
- **Not a causal claim:** This is descriptive out-of-sample variation. No single institutional break is identified.

What this implies about mobilization:

Temporal Predictive Drift by Model Architecture (PR-AUC)



Threshold sensitivity appendix: performance fluctuates

# Qualitative Context: What the Algorithm Cannot See

## The “Motive Blindness” Problem

- The algorithm **cannot distinguish between**:
  - *Exclusionary NIMBYism* (affluent West Austin blocking density)
  - *Anti-displacement resistance* (East Austin protecting vulnerable tenants)
- Same quantitative risk weight; opposite social meaning.

## Practitioner Perspectives (IRB Protocol ACYY0820, $N = 8$ )

- *“Haters are overrepresented and silent supporters are underrepresented.”* (Beaudet)
- Rigid cutoffs fail to *“identify where people that are in actual need actually are.”* (Torrado)

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Seminar 6: Predictive Modeling and Algorithmic Tools

### Predicting NIMBYism in Austin's Housing Reform Era

#### Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

#### Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

#### Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

#### Part 2: Experiences with Neighborhood Opposition

1. When you think of “neighborhood opposition” to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

#### Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

#### Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are experimenting with tools that use data to predict where opposition to development is likely to occur. How do you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).



# Limitations

## Methodological

- **Selection bias / chilling effects:** Only cases that entered the formal process are observed. Developers who anticipated opposition and withdrew or heavily modified proposals are invisible to the model. Attempted IPW failed overlap diagnostics and was discarded.
- **Timestamp imputation:** Milestone dates back-calculated from statutory minimums introduce measurement error in the feature matrix.
- **Unobserved clerk certification:** The outcome is a *reconstructed* area share, not the clerk's legal determination.

## Ethical and Governance

- **Proxy discrimination:** Even without protected-class inputs, reliance on neighborhood income and demographics encodes class and race dynamics — a permanent methodological limitation, not a solved design feature.
- **Motive blindness:** Cannot separate exclusionary NIMBYism from legitimate anti-displacement resistance.
- **External validity:** Parameters are Austin-specific and bounded to the pre-HB 24 policy window. Methodology is portable; specific weights are not.

# Conclusion

## What we showed

- **Institutional Stakes:** Crossing the 20% threshold adds 42.5 days (6.1 weeks) of council processing time — the outcome is worth predicting.
- **Predictive Signal:** Filing-date data ranks petition risk above chance (PR-AUC = 0.718, 95% CI: [0.48, 0.91]), with top-decile lift of 9.9x.
- **Architecture Dependence:** Algorithm choice determines which urban feature clusters drive predictions — a normative, not just technical, decision.

## What surprised us

- **Signal Dilution Paradox:** Retraining on modern data degrades lift; legacy models preserve spatial sorting intelligence learned during an active institutional era.
- **Neighborhood Agency:** Performance ceilings and erratic threshold sensitivity are *consistent with* mobilization being a **strategic social choice**, not a mechanical administrative response.

## What it implies

- Predictions belong in **relative triage rankings**, not case-level probability thresholds.
- Methodology (temporal holdouts, semantic clustering, qualitative integration) is **portable**; Austin-specific parameters are not.

# Implications: A Now-Closed Policy Window

## The HB 24 Shift (September 1, 2025)

- Texas HB 24 exempted residential upzonings and comprehensive mapping efforts from the supermajority petition requirement.
- The 20% institutional veto — as studied here — no longer applies to the most contested case types.
- This thesis maps a specific, now-closed historical era of minority veto power.

## What travels forward

- The **methodology** — time-stamped features, temporal holdout evaluation, semantic clustering, and qualitative integration — is portable to other discretionary review systems.
- Other states retain comparable architectures: Iowa, Oklahoma, Arizona. The analytical framework applies wherever institutional veto mechanisms create predictable opposition patterns.

*This model identifies risk in a historical architecture: current Texas policy has moved*

# Discussion / Q&A

*Thank you to the committee for your time and feedback.*

Dory Kornfeld  
Weiping Wu  
James Piacentini  
Josh Begley

Stijn Van Nieuwerburgh  
Matthew Bauer  
Adam Vosburgh